

Learning and Behavior

Learning & Behavior (PSY/NSC 3320)

Jean P. Bodet III, Ph.D.

Baylor University

THE ROAD SO FAR

An examination of human and animal learning and behavior
within a nature-nurture framework.

NOW

Beyond Behaviorism
The Cognition of Learning

An examination of human and animal learning and behavior
within a nature-nurture framework.

Beyond behaviorism

Behavior vs. Cognition

The Connectionist Model

The Connectionist Model

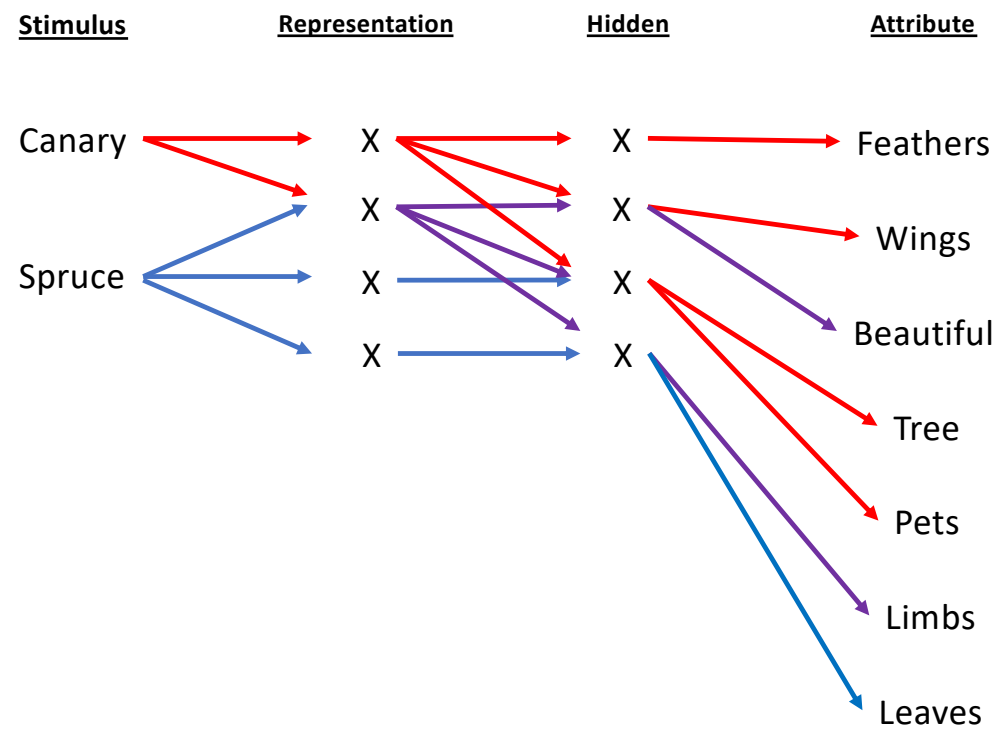
Learning semantic relationships (facts), for instance...

Bird (can) fly
Dog (has) fur

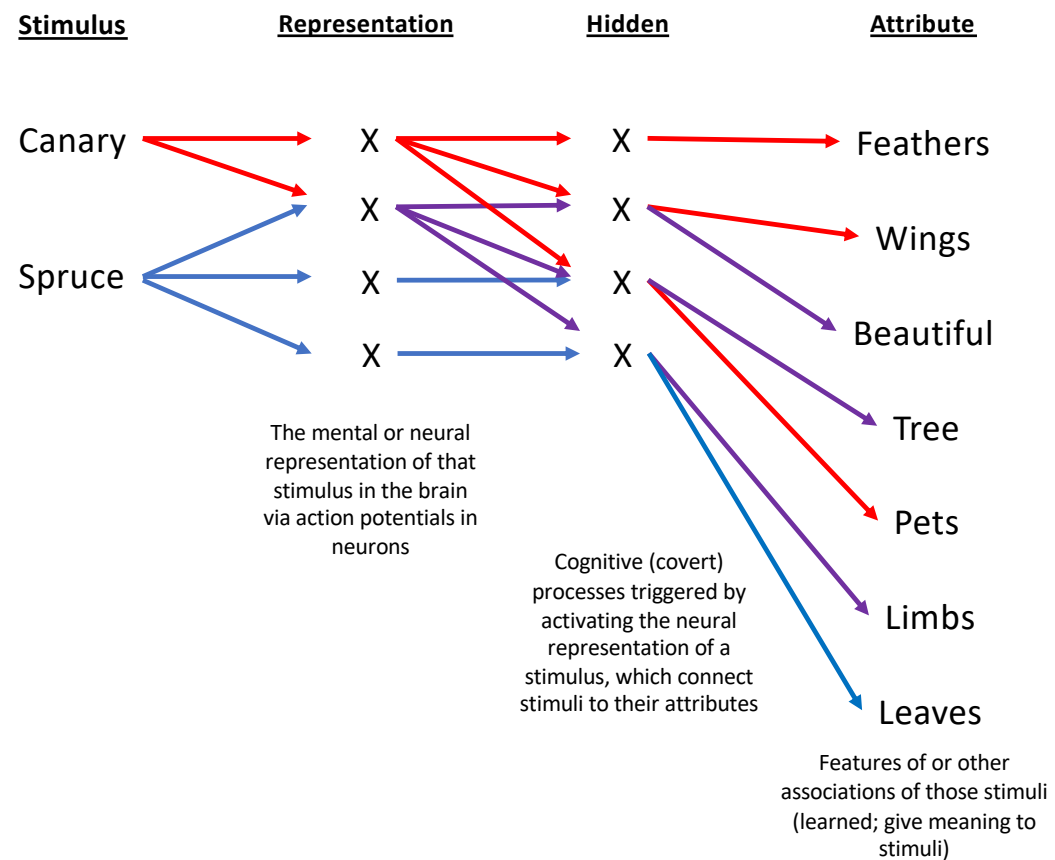
Associative learning of two stimuli, for instance...
CS : US
R -> S

*A representation of the cognition of **learning**, a process that is difficult to observe directly*

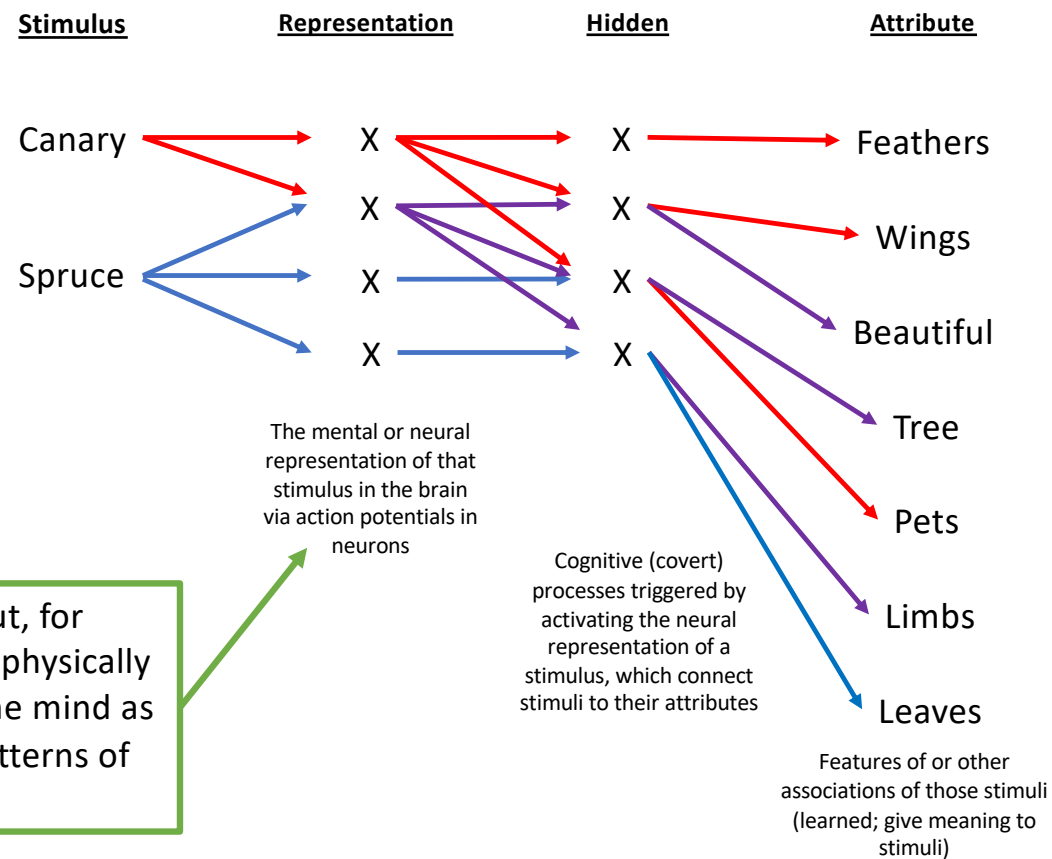
The Connectionist Model



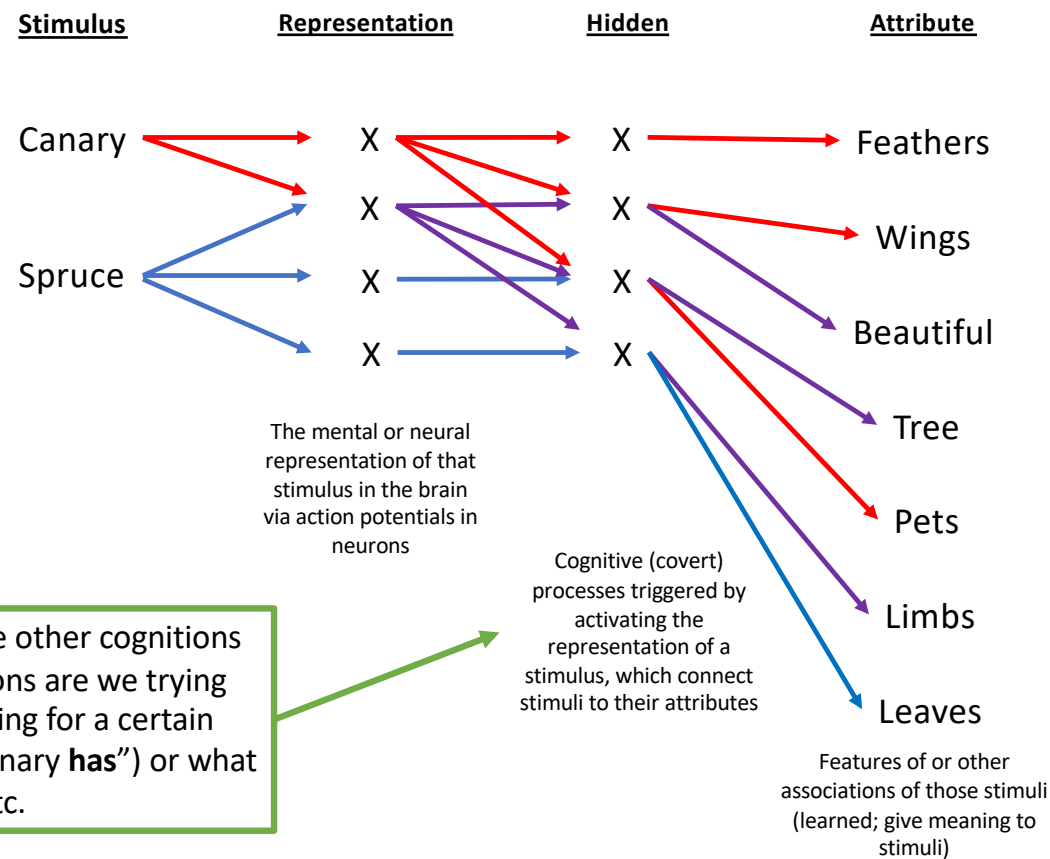
The Connectionist Model



The Connectionist Model

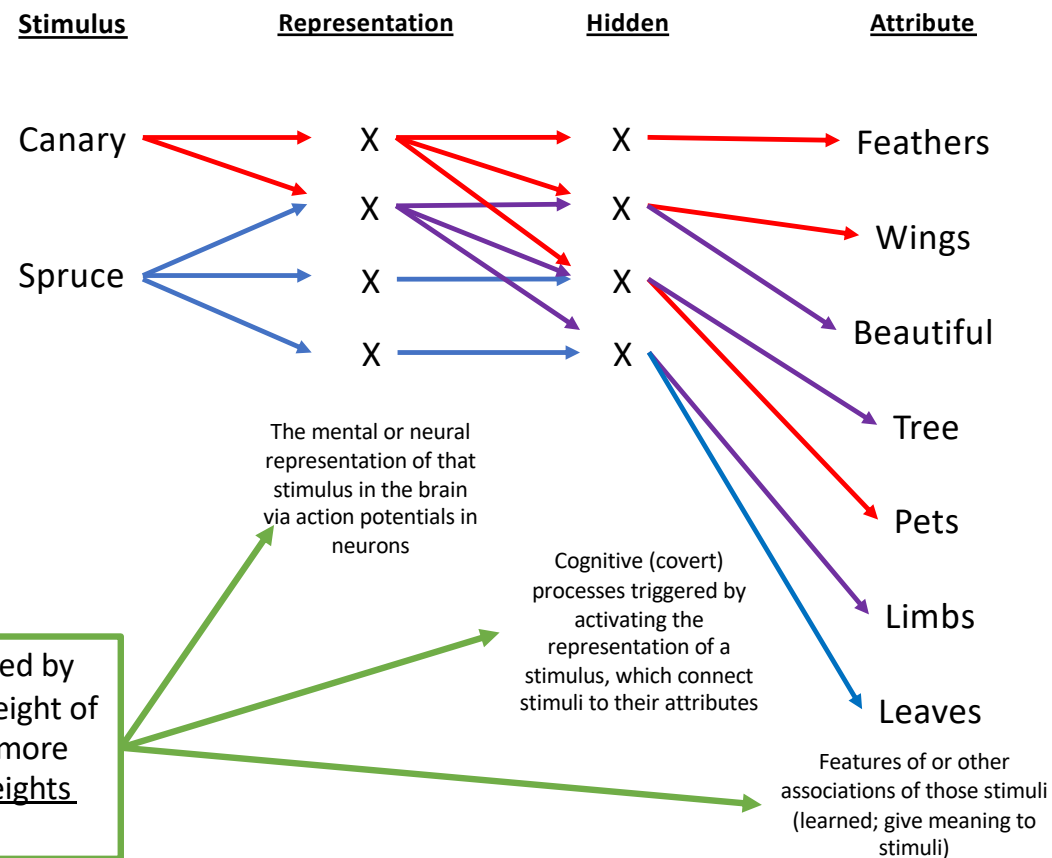


The Connectionist Model

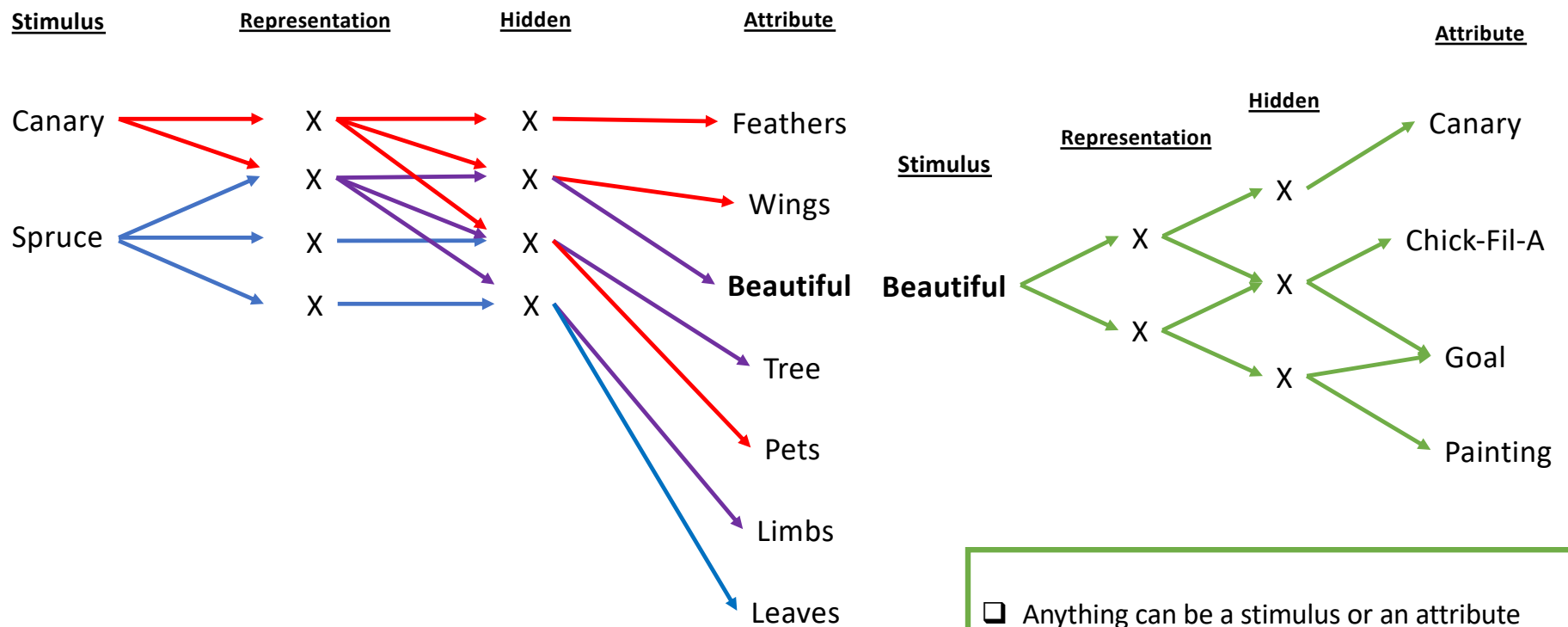


In a **cognitive** sense, "hidden" may be other cognitions like goals (i.e., what kind of associations are we trying to remember, for instance, if I'm looking for a certain type of relationship "canary **is**" or "canary **has**") or what a person is feeling at that moment, etc.

The Connectionist Model



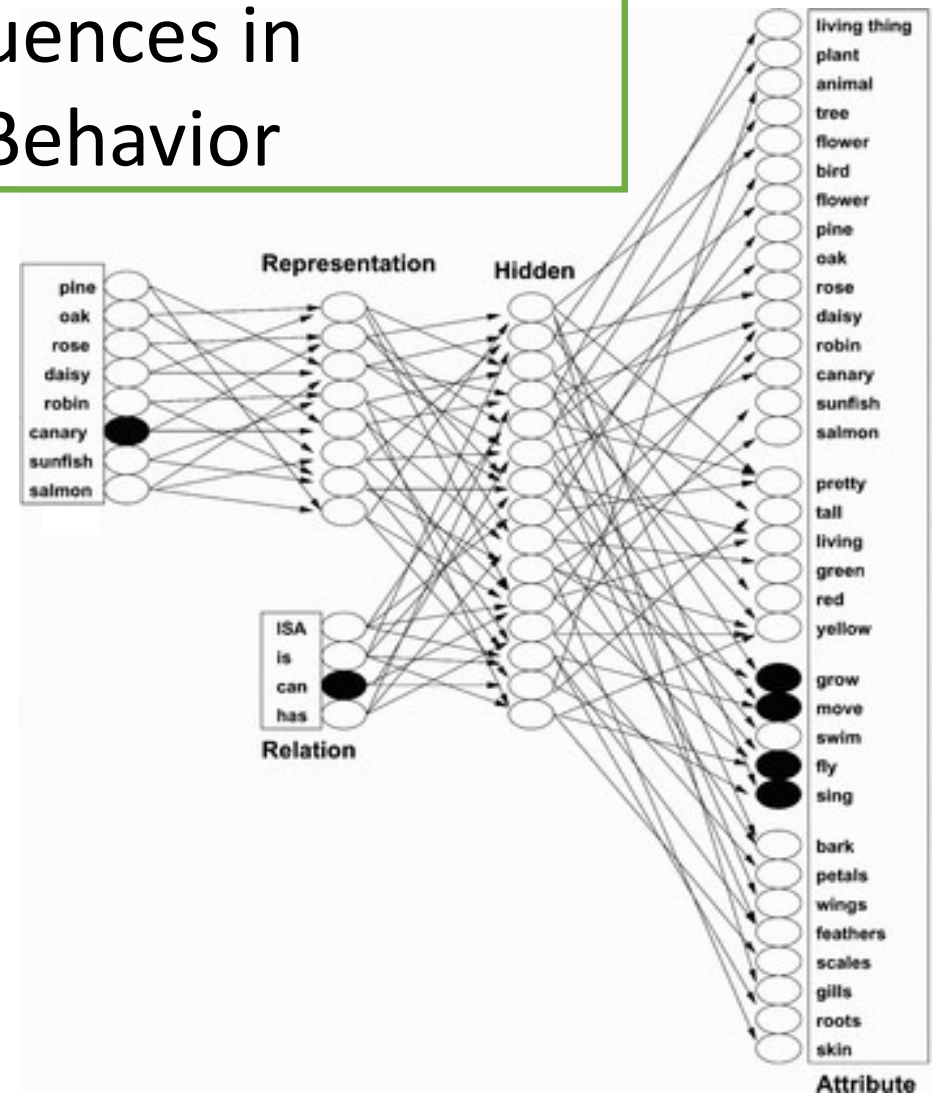
The Connectionist Model



- ☐ Anything can be a stimulus or an attribute
- ☐ This model simply visualizes how the brain organizes associated knowledge

Covert Influences in Learning/Behavior

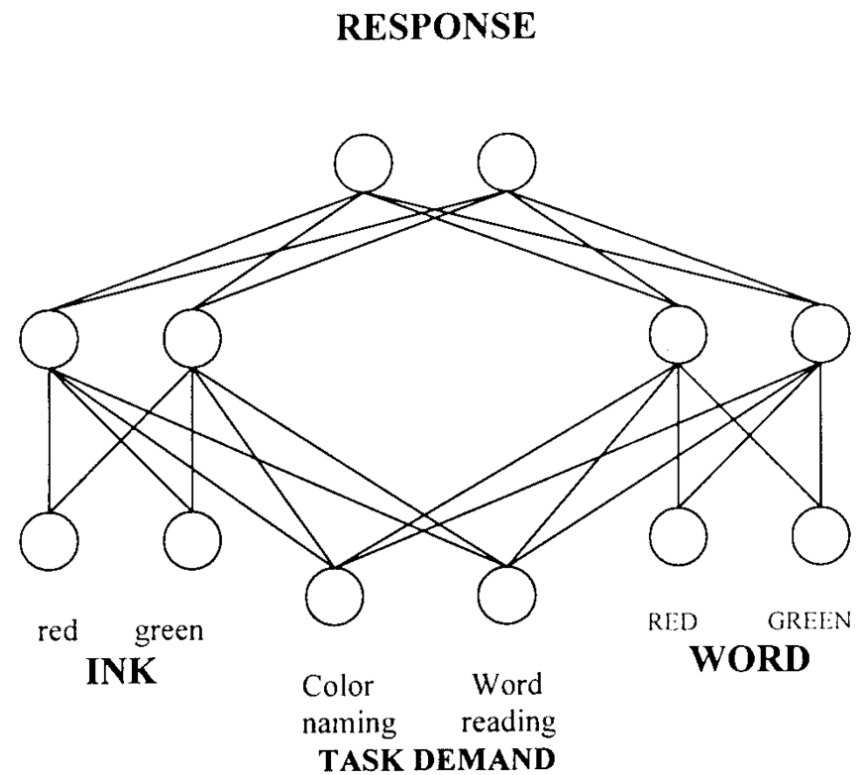
An example from a publication showing this concept in practice



Rumelhart, 1990; Rogers & McClelland, 2004; McClelland 2013

Covert Influences in Learning/Behavior

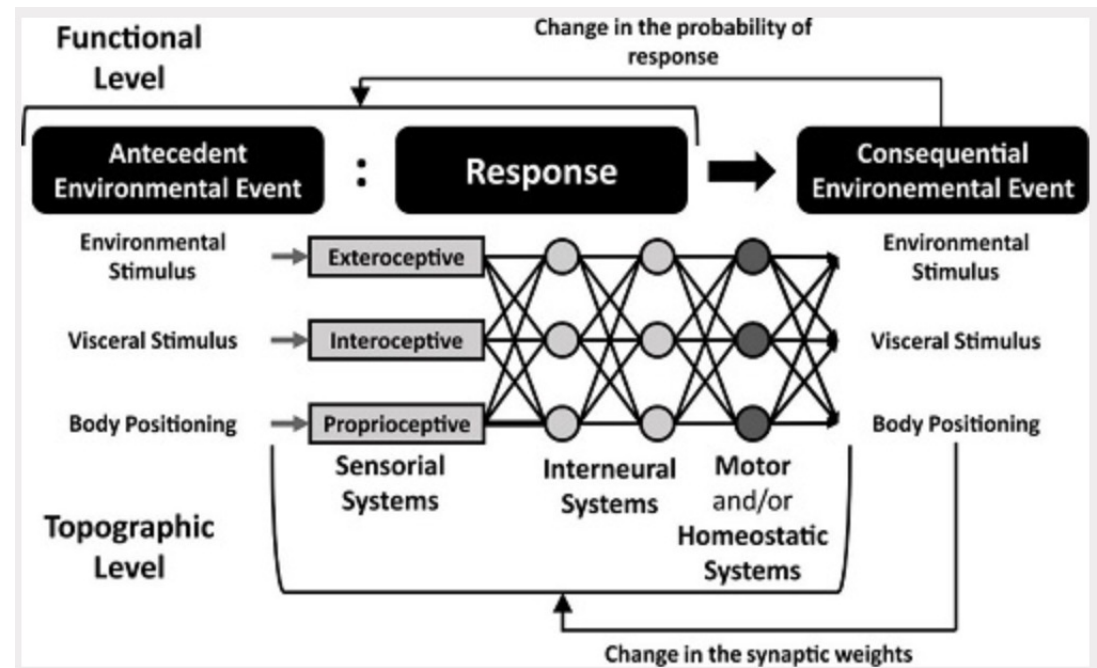
An example applied to processing a very familiar and simple task: Stroop



(Cohen et al., 1990)

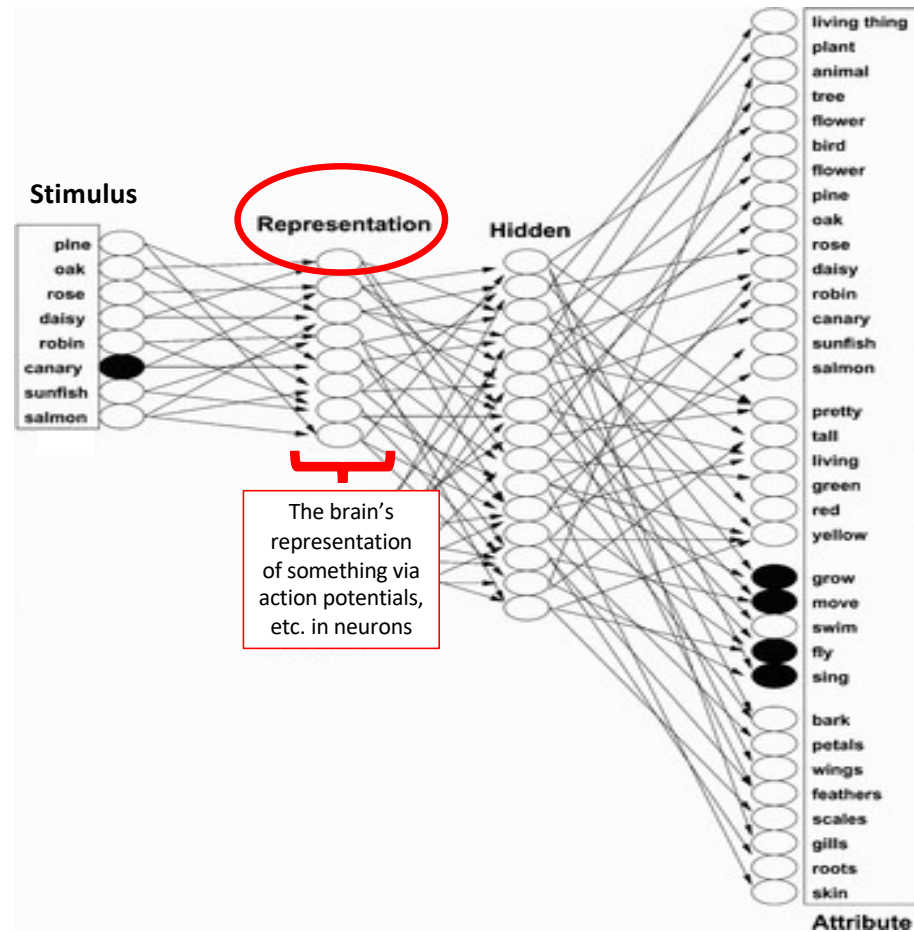
Covert Influences in Learning/Behavior

An example from a publication showing this concept visualizing a behaviorist learning paradigm



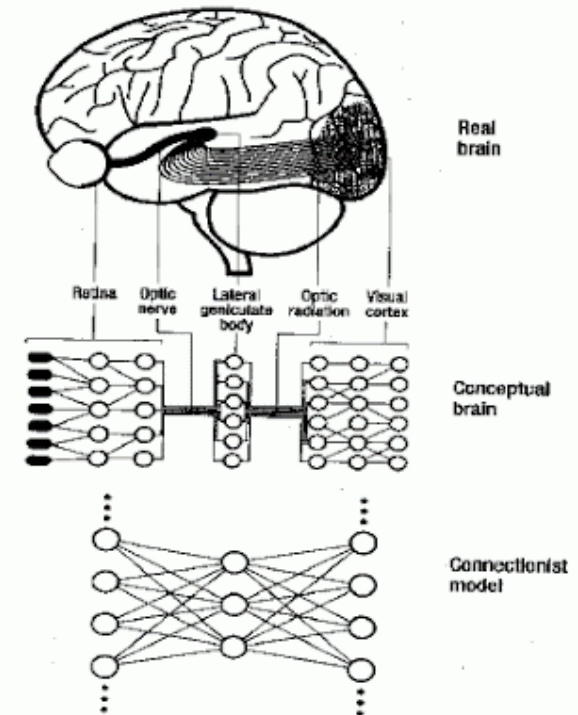
"neural" "hidden" "attributes"
(in this case, motor responses or "B" process stuff)

Key Concept - Neural Representation



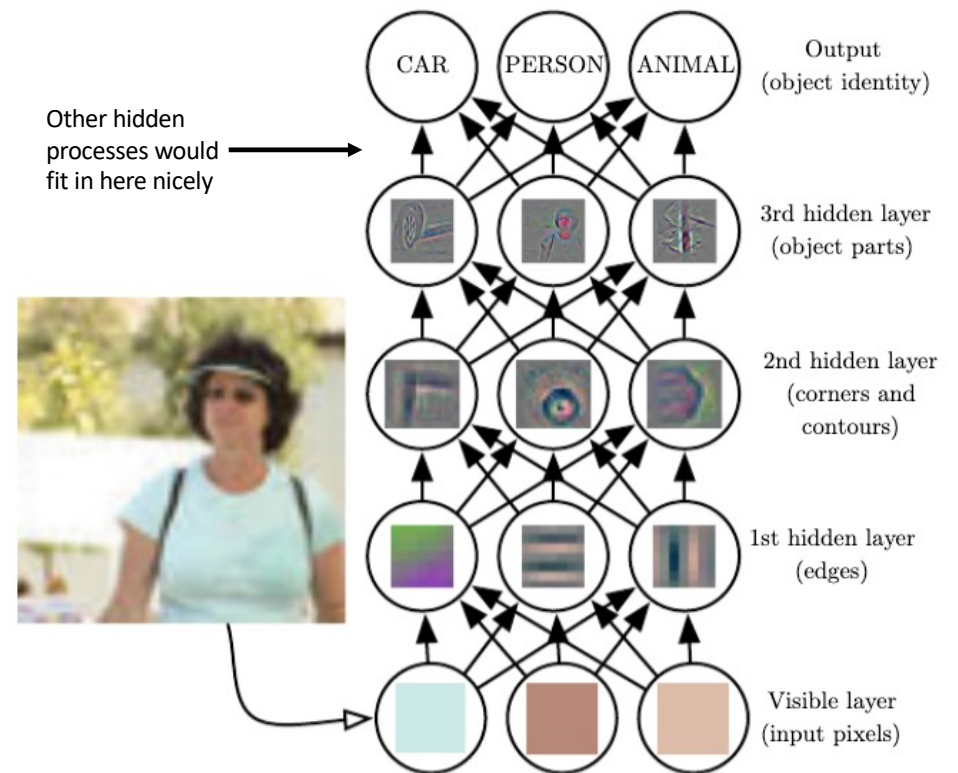
Key Concept - Neural Representation

Figure 4. The layered structure of information processing in the brain and connectionist models. Top: processing of visual information in the brain from retina to lateral geniculate body to visual cortex. Centre: this flow conceptualized as transmission of information through a series of processing layers. Bottom: a connectionist model with three layers of processing units. (Figure taken from McLeod et al., 1998)



Key Concept - Neural Representation

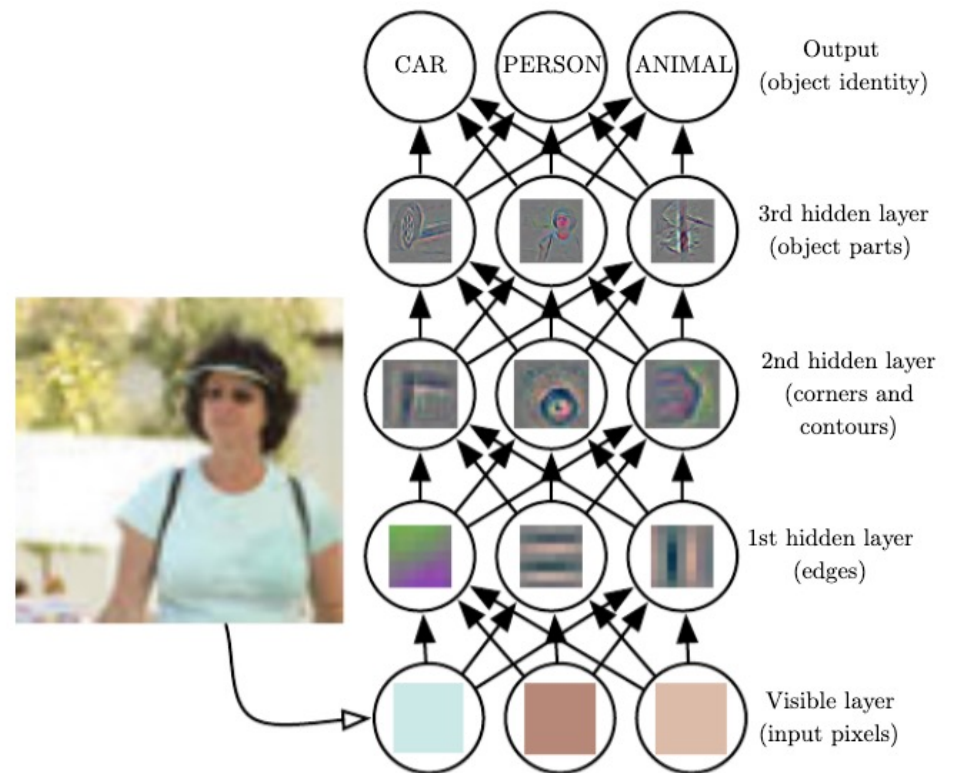
A connectionist model showing the organization of (and different levels of) a visual stimulus' neural representation



Key Concept - Neural Representation

Representing “meaningful” info
that we’ve gleaned from sensory
data

Representing raw sensory data
before we make sense of it

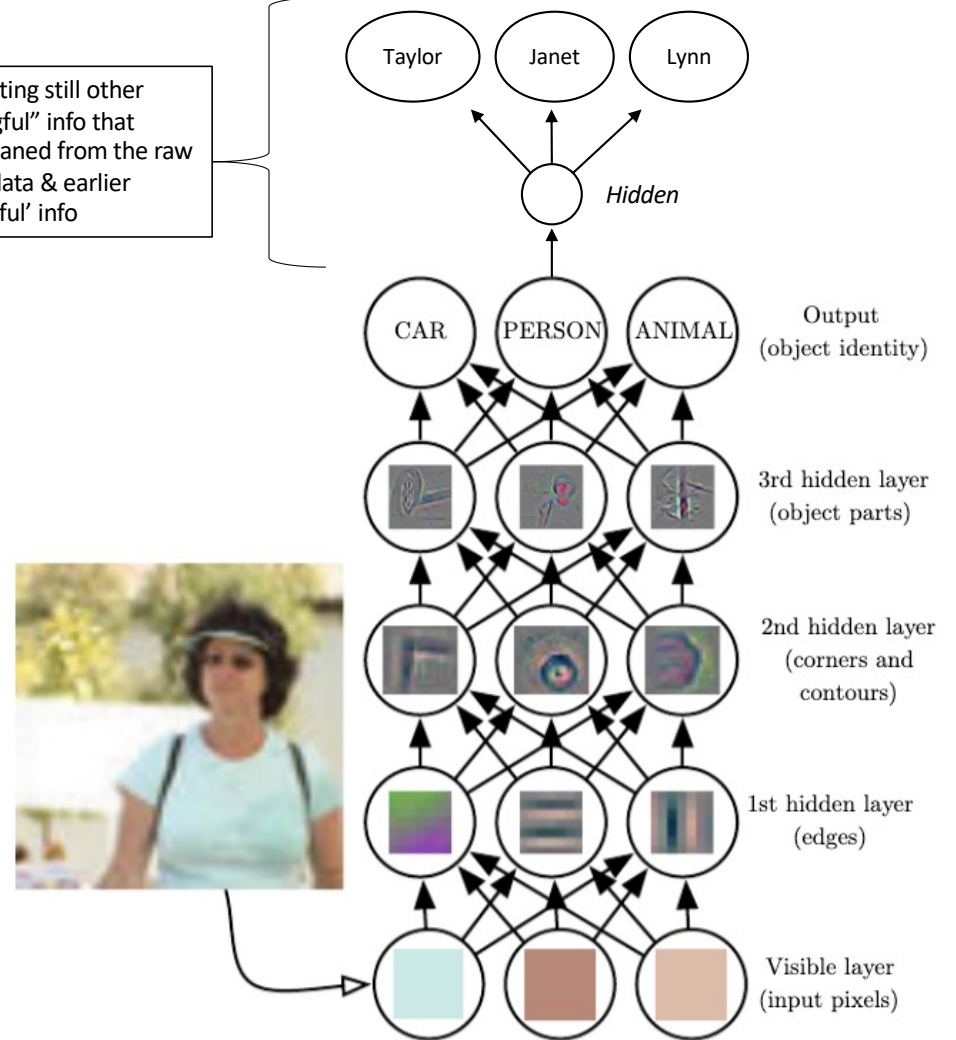


Key Concept - Neural Representation

Representing “meaningful” info that we’ve gleaned from sensory data

Representing raw sensory data before we make sense of it

Representing still other “meaningful” info that we’ve gleaned from the raw sensory data & earlier ‘meaningful’ info

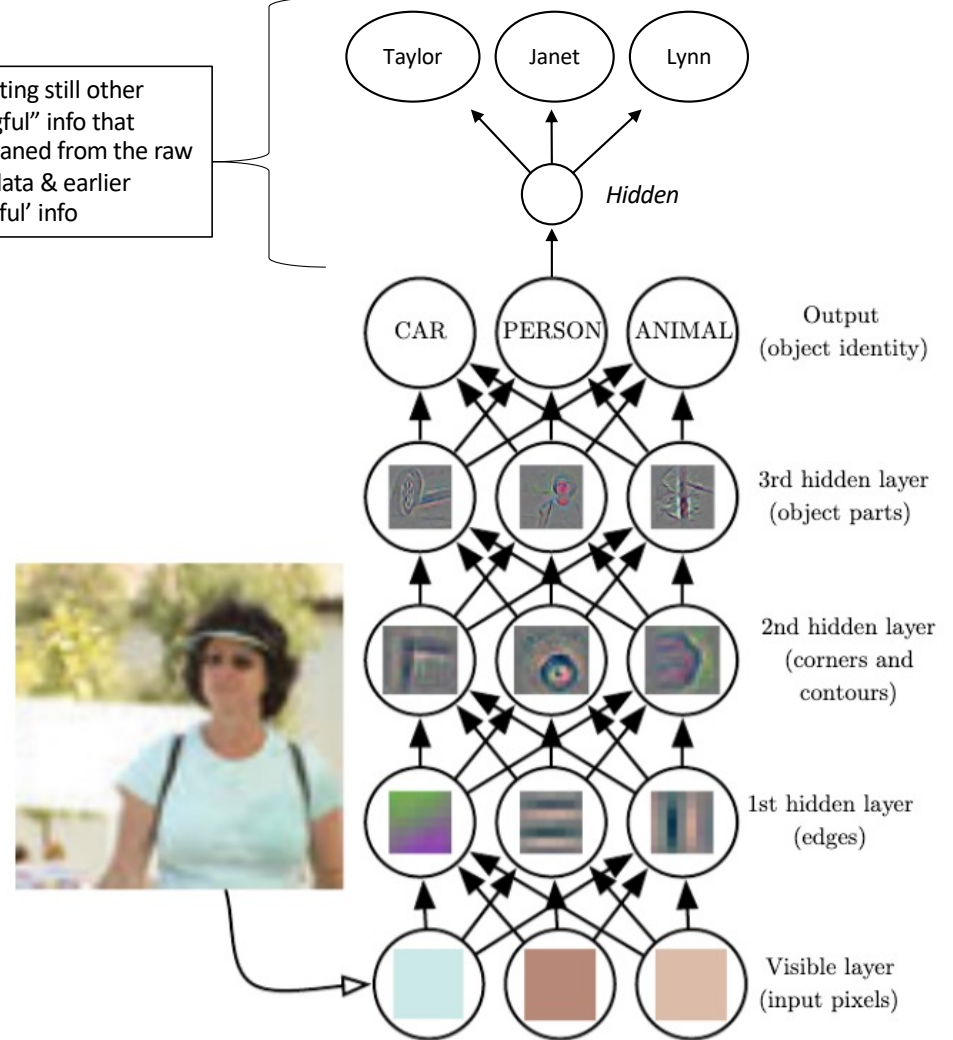


Key Concept - Neural Representation

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Biologically, long-term memory might be described as...

Preserving a record of which neurons represented some experience (*i.e.*, were involved in processing/making sense of that experience)

Covert Influences in Learning/Behavior

Connectionist Model

We can take this “Connectionist” approach to learning and apply it to many types of memory; we’ve just seen it applied to *associative* (a form of implicit) and *semantic* (a form of explicit) memories

Questions?